

Superstore Sales Analysis Report

This report presents a comprehensive analysis of Superstore sales data using Python and Power BI. The project includes data cleaning, exploratory data analysis, visualization, and dashboard development to identify business trends, profitable categories, and strategic recommendations.

Project Objectives

- Clean and preprocess raw sales data
- Analyze sales and profit trends
- Identify top-selling products
- Analyze regional and category performance
- Build an interactive Power BI dashboard
- Generate actionable business insights and recommendations

Dataset Information

The dataset used in this project is the Sample Superstore dataset containing sales transactions, customer information, regional data, and profitability metrics.

Data Cleaning & Preprocessing

- Removed duplicate records
- Checked and handled missing values
- Converted order and ship dates into datetime format
- Created additional columns such as Year and Month
- Verified data types and consistency

Key Business Metrics

Metric	Value
Total Sales	\$2,297,200
Total Profit	\$286,397
Best Category	Technology
Lowest Profit Category	Furniture
Top Region	West
Lowest Region	South

Key Insights

- Technology category generated the highest profit and sales.
- Furniture category showed high sales but low profitability.

- West region contributed the highest revenue.
- Business growth accelerated significantly after 2015.
- A small number of products contributed major revenue.

Business Recommendations

- Focus on promoting high-margin technology products.
- Reduce excessive discounts on furniture products.
- Improve marketing campaigns in underperforming regions.
- Optimize shipping and logistics costs.
- Use customer segmentation for targeted marketing.

Power BI Dashboard Features

- Interactive KPI Cards
- Sales Trend Analysis
- Category-wise Sales Visualization
- Region-wise Performance Analysis
- Top Products Analysis
- Interactive Filters and Slicers

Conclusion

This project demonstrates practical skills in data cleaning, exploratory data analysis, business intelligence, and dashboard creation. The analysis provides valuable business insights that can support strategic decision-making and operational improvements.

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